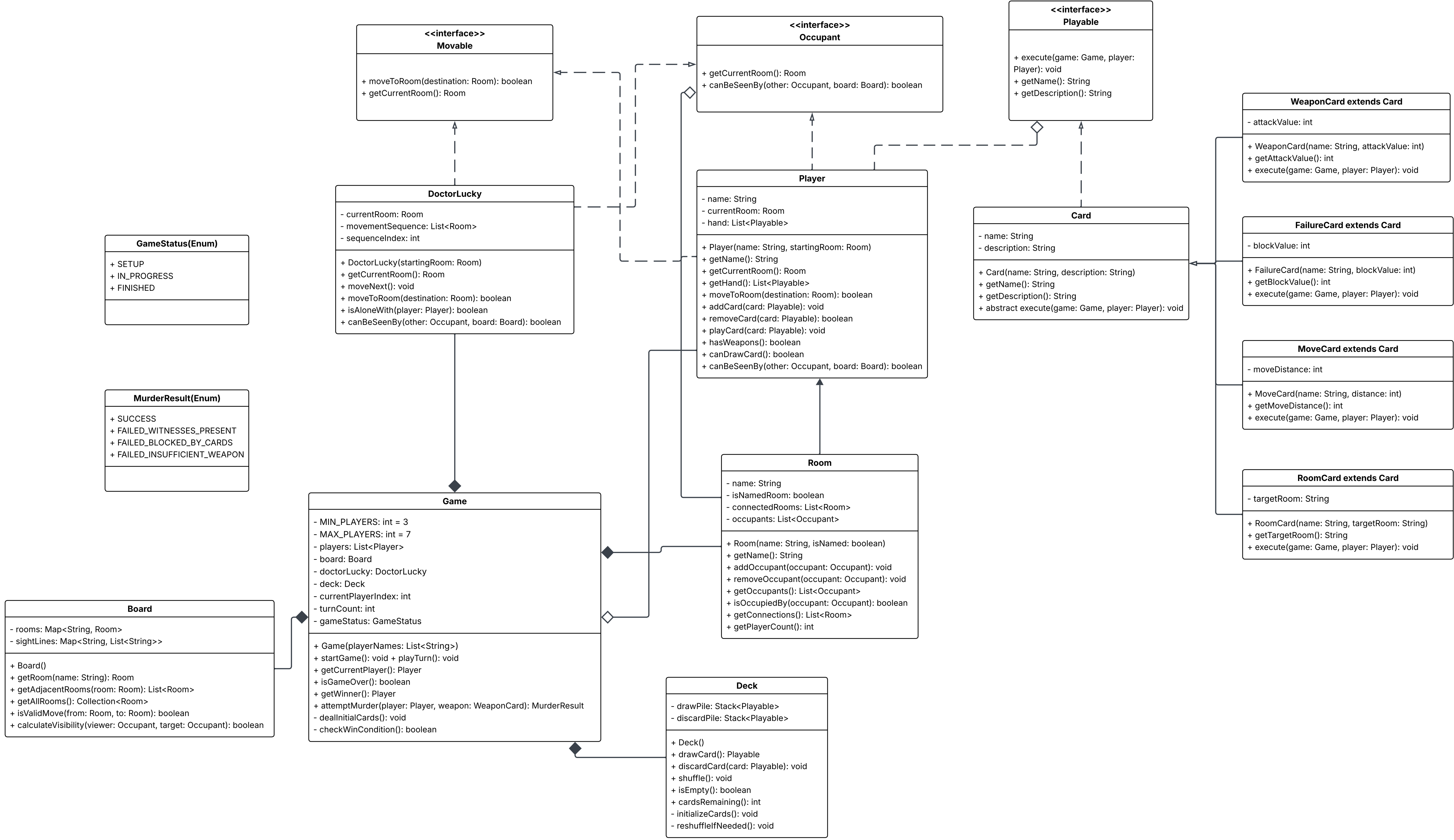




Test Plan for KillDoctorLucky

1. Game Class Test

a. Valid game initialization

- Condition: Game constructor with valid player names
 - Example Data: ["Peter", "Will", "Alice", "Henry"]
 - Expected: the Game created, gameStatus = SETUP
- Minimum Players (Valid)
 - Condition: Game constructor with minimum players to be 3
 - Example Data: ["Peter", "Will", "Alice"]
 - Expected: Game created successfully.
- Maximum Players (Valid)
 - Condition: Game constructor with maximum players to be 7
 - Example Data: ["Peter", "Will", "Alice", "Henry", "Alan", "Olivia", "George"]
 - Expected: Game created successfully.
- Below Minimum Players
 - Condition: Game constructor with 2 players
 - Example: ["Peter", "Henry"]
 - Expected: Exception thrown
- Above Maximum Players
 - Condition: Game constructor with 8 players
 - Example: ["Peter", "Will", "Alice", "Henry", "Alan", "Olivia", "George", "Jason"]
 - Expected: Exception thrown

f. Start Game

- Condition: startGame called
- Example: Game with 3 players
- Expected: SETUP = IN_PROGRESS, cards dealt to players

g. Successful Murder

- Condition: attemptMurder() with player alone, strong weapon
- Example: Alice alone with Doctor Lucky, WeaponCard(value = 10)
- Expected: MurderResult.SUCCESS, game over.

h. Failed Murder - Witness

- Condition: attemptMurder() with witness present
- Example: Alice, Henry and Doctor Lucky all in same room
- Expected: MurderResult.FAILED_WITNESS_PRESENT

2. Player Class Test

a. Player Creation

- Condition: Player constructor with valid data
- Example: name = "Alice", startingRoom = Kitchen
- Expected: Player created, currentRoom = Kitchen, empty hand

b. Valid Movement

- Condition: moveToRoom() to connected room
- Example: Player in Kitchen, move to Dining Room
- Expected: Returns true, currentRoom = Dining Room

c. Invalid Movement

- Condition: moveToRoom() to unconnected room
- Example: Player in Kitchen, move to Bedroom (not connected)
- Expected: Return False, currentRoom unchanged

d. Add Card to Hand

- Condition: addCard() with space available
- Example: Player with 3 cards, add WeaponCard("Knife", 5)
- Expected: hand. size() = 4, card added

e. Hands initialization

- Condition: Player starts with 6 cards
- Example: New Game initialization
- Expected: each player's hand.size() = 6

3. Board and Room Test

a. Get Valid Room

- Condition: getRoom() with existing room name
- Example: getRoom("Kitchen")
- Expected: Returns Kitchen room object

b. Invalid Room

- Condition: getRoom() with non-existent room
- Example: getRoom("InvalidRoom")
- Expected: Returns null or throws exception

c. Add Occupant to Room

- Condition: addOccupant() with player
- Example: add Henry to Kitchen
- Expected: Alice in Kitchen's occupant list

d. Room Visibility

- Condition: calculateVisibility() between rooms
- Example: Alice in Kitchen and Bob in Dining Room (connected)
- Expected: Returns True (can see each other)

4. Doctor Lucky Test

a. Sequential Movement

- Condition: moveNext() called multiple times
- Example: Movement sequence [Kitchen, Dining, Living], call 3 times
- Expected: Kitchen to Dining to Living to Kitchen (circle)

b. Alone Check-True

- Condition: isAloneWith() with only player present
- Example: Kitchen has only Alice and Doctor Lucky
- Expected: isAloneWith() return True

c. Alone Check-False

- Condition: isAloneWith() with multiple player
- Example: Kitchen has Alice, Henry and Doctor Lucky
- Expected: isAloneWith() return False

5. Card Test

a. Weapon Card

- Condition: WeaponCard creation and execution
- Example: WeaponCard("Knife", 5)
- Expected: Card created attackValue = 5

b. Failure Card

- Condition: FailureCard blocks murder attempt
- Example: FailureCard("Bad Luck", 3) used during murder
- Expected: reduce murder success by 3 points

c. Move Card

- Condition: MoveCard allows extra movement
- Example: MoveCard("Quick Step", 2)
- Expected: Player can move up to 2 additional rooms

d. Room Card

- Condition: RoomCard instant teleportation
- Example: RoomCard("Kitchen Card", "Kitchen")
- Expected: player instantly move to Kitchen

6. Deck Test

a. Empty Deck Reshuffle

- Condition: drawCard() when draw pile empty
- Example: Empty draw pile, 20 cards in discard
- Expected: discard reshuffled, card drawn

b. Draw Card

- Condition: drawCard() with cards available
- Example: Deck with 50 cards
- Expected: return card, deck size decrease

c. Discard Card

- Condition: discardCard() adds to discard pile
- Example: discard WeaponCard("Knife", 5)
- Expected: Card added to discard pile

7. Interface Test

a. Playable Interface

- Condition: All cards implement Playable
- Example: WeaponCard, MoveCard treated as Playable
- Expected: execute(), getName() work for all cards

b. Movable Interface

- Condition: Player and Doctor Lucky implement Movable
- Example: Both objects as Movable type
- Expected: moveToRoom(), getCurrentRoom() work for both

8. Integration Test

a. Game end

- Condition: Game ends when murder successful
- Example: Alice successfully murders Doctor Lucky
- Expected: gameStatus = FINISHED, Alice declared winner

b. Multiple player interaction

- Condition: players interact during murder attempt
- Example: Alice attempts murder, Bob plays FailureCard
- Expected: Bob's card affects Alice's murder attempt

c. Complete Turn

- Condition: full player turn with movement and card play
- Example: Alice moves, plays MoveCard, attempts murder
- Expected: all actions execute correctly, turn passes to next player